

KIEL HOLTENAU, Germany

WMO: 100460

Lat: 54.3761N

Lon: 10.1434E

Elev: 97

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 95-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	21.2	12.5	10.4	19.7	16.1	12.5	23.4	30.1	42.0	27.3	43.8	6.2	270	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.8	81.0	66.4	77.3	65.0	73.8	63.4	68.3	78.1	66.5	75.1	64.8	72.5	8.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	91.6	72.1	62.9	86.0	70.6	61.2	80.9	68.6	32.5	78.4	31.0	75.2	29.8	72.4	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.9	17.6	DB	12.3	88.6	4.4	3.1	9.1	90.8	6.5	92.6	4.0	94.3	0.8	96.6
			WB	12.1	71.9	4.3	1.4	8.9	73.0	6.4	73.8	4.0	74.6	0.8	75.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	35.0	35.7	40.0	47.0	53.5	58.9	63.5	63.9	58.1	49.9	42.4	36.0
	DBStd	12.06	7.39	7.31	6.51	6.10	5.47	5.40	5.18	4.95	5.09	5.66	6.14	7.19
	HDD50	2074	464	401	312	129	27	1	0	0	1	72	234	433
	HDD65	6056	929	820	774	542	360	197	93	82	216	468	677	898
	CDD50	1612	0	1	3	38	134	268	419	430	244	68	7	0
	CDD65	119	0	0	0	0	2	14	47	47	9	0	0	0
	CDH74	785	0	0	0	6	25	117	320	275	42	0	0	0
	CDH80	154	0	0	0	0	5	28	64	54	3	0	0	0
Wind	WSAvg	8.7	9.9	10.3	9.7	8.3	8.3	8.5	8.1	7.5	7.9	8.5	8.8	9.3
Precipitation	PrecAvg	30.7	2.2	1.8	1.7	2.2	2.0	2.5	3.3	3.2	2.6	2.5	3.7	3.0
	PrecMax	37.3	4.6	2.9	2.8	5.0	3.1	4.2	6.9	8.3	3.9	4.0	8.0	6.3
	PrecMin	21.6	0.5	0.5	0.4	0.5	0.7	1.2	1.1	1.4	0.6	0.4	1.3	0.4
	PrecStd	5.0	1.2	0.8	0.7	1.2	0.8	0.9	1.6	2.1	0.8	1.1	1.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	53.7	61.0	71.9	77.1	84.2	86.2	84.6	78.9	66.5	57.6	52.2
		MCWB	48.5	48.3	50.5	57.9	62.2	68.5	67.8	67.6	64.3	59.4	53.7	49.2
	2%	DB	49.7	50.2	55.5	64.7	70.5	77.0	80.9	80.7	73.7	62.7	55.1	49.9
		MCWB	47.0	46.7	48.4	53.6	58.4	64.4	66.4	66.8	62.8	57.5	51.7	47.8
	5%	DB	46.7	47.9	52.0	61.0	66.6	72.7	77.2	76.8	69.9	60.4	52.2	46.8
		MCWB	44.6	45.6	46.8	52.2	56.6	62.2	65.5	65.4	61.3	55.9	49.1	45.2
	10%	DB	44.7	45.0	49.6	57.3	64.2	68.4	73.7	73.4	66.4	57.6	50.3	45.0
		MCWB	42.8	43.2	45.5	49.9	55.3	59.9	63.6	63.9	59.6	53.9	47.9	43.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	49.8	52.3	58.6	63.5	69.6	70.6	70.5	66.8	60.7	55.0	50.8
		MCDB	51.4	52.0	57.4	68.8	74.0	80.7	79.9	80.3	74.9	65.6	57.4	52.6
	2%	WB	46.7	47.2	49.8	55.4	60.2	65.8	68.5	68.0	64.1	58.0	52.1	47.8
		MCDB	48.5	49.2	54.3	63.3	67.8	75.2	78.2	77.5	71.4	61.8	55.1	49.4
	5%	WB	44.6	45.3	47.6	52.9	58.1	63.2	66.5	66.2	62.1	56.3	50.2	45.5
		MCDB	46.4	47.1	51.7	60.1	65.0	70.1	75.2	74.4	68.8	59.7	52.2	46.9
	10%	WB	42.6	43.2	45.3	50.6	56.0	60.9	64.4	64.9	60.2	54.7	48.4	43.5
		MCDB	44.2	44.9	48.3	56.1	62.3	67.5	72.2	72.2	65.8	57.9	50.3	44.8
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.9	11.3	14.7	16.0	15.6	15.2	14.8	13.2	10.7	7.7	6.9
		MCDBR	8.1	8.9	15.9	21.0	21.3	20.9	20.7	19.7	18.2	13.2	8.5	7.6
		MCWBR	7.5	7.4	10.9	12.5	12.6	11.7	10.2	10.2	10.1	9.3	7.3	7.3
	5% WB	MCDBR	7.8	8.4	14.6	19.1	18.8	18.8	19.4	17.1	16.2	11.7	8.4	7.6
		MCWBR	7.4	7.4	10.8	12.0	12.3	11.3	10.6	9.9	9.9	8.9	7.5	7.5
Clear-Sky Solar Irradiance	taub	0.318	0.339	0.367	0.406	0.398	0.395	0.413	0.410	0.384	0.358	0.335	0.308	
	taud	2.301	2.319	2.269	2.209	2.306	2.343	2.308	2.329	2.383	2.418	2.361	2.295	
	Ebn at Noon	197	235	254	259	269	271	263	256	247	226	190	174	
	Edh at Noon	18	25	33	40	38	37	38	35	29	23	17	15	
All-Sky Solar Radiation	RadAvg	160	361	768	1302	1655	1740	1630	1320	952	504	206	116	
	RadStd	22	56	72	155	179	127	193	96	99	62	25	8	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-268	N/A	N/A

Nomenclature: See separate page